

## MATHEMATICAL BIOLOGY

# Superinfections Can Induce Evolutionarily Stable Coexistence of Pathogens

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*Student:*  
Hengyi Li

*Lecturers:*  
Dr. S. C. Hille  
Dr. R. Planqué

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## Abstract

This essay introduces a new infection pattern: *superinfection*, meaning reinfection of already infected hosts, to combine two scales of natural selection. We analyse the models and find that superinfection can imply higher virulence and yield evolutionarily stable coexistence of pathogen strains.

## 1 Introduction

The evolution of pathogens is influenced by natural selection across multiple scales. For many microparasites (viruses, bacteria, and protozoa), replication occurs *within host*, yet long-term survival requires successful transmission *between hosts*. However, traits that maximise competitive advantage within the host may reduce host survival or alter transmission. These two levels of natural selection may therefore favour different evolutionarily stable traits [1].

We assume pathogen strains compete for a single limited resource: the *target cells*. By the *pessimisation principle*,<sup>1</sup> the optimal *reproduction rate*  $p$  minimises the availability of uninfected

<sup>1</sup>If the population growth rate is a monotone increasing function of a one-dimensional variable that fully describes the environmental condition, then the evolutionary winner is the trait that *minimises* this variable.

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target cells. At the host-population level, pathogen strains compete for susceptible hosts via transmission, which can be modelled by a simple susceptible-infected (SI) model:

$$\begin{aligned}\frac{dS}{dt} &= b - \delta S - \beta SI, \\ \frac{dI}{dt} &= -\delta I + \beta SI - \alpha I,\end{aligned}$$

where  $b$  denotes population birth rate;  $\alpha$  is the *virulence*;  $\delta$  denotes death rate related to other causes; and  $\beta$  is the transmission rate (assumed by  $\beta = \beta(\alpha)$ ).

In nested models, however,  $\alpha$  and  $\beta$  are not free parameters but mechanistically linked to within-host equilibrium quantities. We resolved the inconsistency by linking the two levels through *superinfection*: when pathogens spread to already infected hosts, new strains may take over the host almost immediately.

In this essay, we shall introduce the models in detail and investigate how permitting superinfection alters the host-internal production rate  $p$  under evolutionary selection. *Can superinfection lead to an evolutionarily stable coexistence of pathogen strains, thereby maintaining polymorphism over evolutionary timescales?*

## 2 Models

In this section, we introduce mathematical models of within-host dynamics and between-host transmission, along with certain preliminaries, thereby laying the groundwork for the superinfection model. We assume that within-host dynamics equilibrate quickly relative to host demography and transmission, and mutations are rare on the ecological timescale.

### 2.1 Within-Host Model

We construct a target-cell-limited model to describe within-host infection dynamics, comprising the following system of ordinary differential equations (ODEs).

**Definition 2.1** (Within-Host Dynamics).

$$\frac{dT}{dt} = \lambda - kVT - dT, \tag{2.1}$$

$$\frac{dT^*}{dt} = kVT - (\mu(p) + d)T^*, \tag{2.2}$$

$$\frac{dV}{dt} = pT^* - kVT - cV, \tag{2.3}$$

where we have such variables:

- $T$ ,  $T^*$ , and  $V$ : The number of uninfected target cells, infected target cells, and free pathogens, respectively.
- $\lambda$  and  $d$ : The production rate and natural mortality of target cells.
- $k$ : The mass action coefficient for infection.  $kVT$  represents the successful encounter and entry of pathogens into target cells.
- $c$ : The clearance rate of free pathogens.
- $p$ : The production rate of free pathogens by infected cells.  $\mu(p)$  is the additional mortality rate of infected cells with increasing  $p$ . We assume that  $\mu(p) \in C^2(\mathbb{R})$  is non-negative and increasing, encoding the biological cost of increased production, and that it is differentiable, as required in later analysis.

*Remark.* A common simplification is to drop the  $-kVT$  term in (2.3), which is effective in certain stages of HIV progression. However, it may not be appropriate for the very first stages. When mutant strains emerge and the pathogen is scarce, a complete model is required. We shall explore the dynamics in the initial stages later in this essay.

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The non-infectious steady state where no pathogens and no infected target cells is:

$$\hat{V} = \hat{T}^* = 0, \quad \hat{T} = \frac{\lambda}{d} =: T_0.$$

Then, we turn our attention to the non-trivial steady state. At the equilibrium, setting  $\dot{T} = \dot{T}^* = \dot{V} = 0$  yields the steady-state system of equations:

$$0 = \lambda - k\hat{V}\hat{T} - d\hat{T}, \quad (2.4)$$

$$0 = k\hat{V}\hat{T} - (\mu(p) + d)\hat{T}^*, \quad (2.5)$$

$$0 = p\hat{T}^* - k\hat{V}\hat{T} - c\hat{V}, \quad (2.6)$$

where we assume that  $\hat{T}^* > 0$  and  $\hat{V} > 0$ . By (2.5), we have

$$\hat{T}^* = \frac{k\hat{V}\hat{T}}{\mu(p) + d}.$$

Combining (2.5) and (2.6),

$$c\hat{V} = (p - (\mu(p) + d))\hat{T}^*. \quad (2.7)$$

Here, we introduce the *burst size*  $\mathcal{B}_0$ , which is the expected number of free pathogens released upon a single infected target cell, defined by

$$\mathcal{B}_0 = \frac{p}{\mu(p) + d}.$$

Then

$$p - (\mu(p) + d) = (\mu(p) + d)(\mathcal{B}_0(p) - 1).$$

Plugging back to (2.7) gives

$$\begin{aligned} c\hat{V} &= (\mu(p) + d)(\mathcal{B}_0(p) - 1)\hat{T}^* \\ &= (\mu(p) + d)(\mathcal{B}_0(p) - 1) \frac{k\hat{V}\hat{T}}{\mu(p) + d} \\ &= (\mathcal{B}_0(p) - 1)k\hat{V}\hat{T}, \end{aligned}$$

which implies

$$c = (\mathcal{B}_0(p) - 1)k\hat{T} \implies \hat{T} = \frac{c}{k(\mathcal{B}_0(p) - 1)}.$$

By (2.7), we have

$$\hat{V} = \frac{\lambda - d\hat{T}}{k\hat{T}} = \frac{\lambda - d \frac{c}{k(\mathcal{B}_0(p) - 1)}}{k \frac{c}{k(\mathcal{B}_0(p) - 1)}} = \frac{\lambda - \frac{cd}{k(\mathcal{B}_0(p) - 1)}}{\frac{c}{(\mathcal{B}_0(p) - 1)}} = \frac{\lambda(\mathcal{B}_0(p) - 1)}{c} - \frac{d}{k}.$$

Combining with (2.4) and (2.5),

$$\hat{T}^* = \frac{\lambda - d\hat{T}}{\mu(p) + d} = \frac{\mathcal{B}_0(p)}{p} (\lambda - d\hat{T}) = \frac{\mathcal{B}_0(p)}{p} \left( \lambda - \frac{cd}{k(\mathcal{B}_0(p) - 1)} \right).$$

Therefore, we conclude a stable infected equilibrium  $(\hat{T}(p), \hat{T}^*(p), \hat{V}(p))$  is given by:

$$\hat{T} = \frac{c}{k(\mathcal{B}_0(p) - 1)}, \quad (2.8)$$

$$\hat{T}^* = \frac{\mathcal{B}_0(p)}{p} \left( \lambda - \frac{cd}{k(\mathcal{B}_0(p) - 1)} \right), \quad (2.9)$$

$$\hat{V} = \frac{\lambda}{c} (\mathcal{B}_0(p) - 1) - \frac{d}{k}. \quad (2.10)$$

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Notice that  $\hat{T} > 0$  requires  $\mathcal{B}_0(p) > 1$ . For  $\hat{T}^* > 0$ , we have

$$\lambda - \frac{cd}{k(\mathcal{B}_0(p) - 1)} > 0,$$

which is equivalent to

$$\mathcal{B}_0(p) > \frac{k\lambda + cd}{k\lambda}.$$

We rewrite the inequality and define the *within-host basic reproduction ratio*:

$$\mathcal{R}_0^w(p) := \frac{k\lambda}{k\lambda + dc} \mathcal{B}_0(p) > 1.$$

Finally, we assume that target cells infected by a single trait are protected from further infection; hence, strains compete for a single resource: uninfected target cells. Then by the *pessimisation principle*, within-host competition tends to minimise  $\hat{T}(p)$ , which is equivalent to maximising  $\mathcal{R}_0^w(p)$ , and also to maximising  $\mathcal{B}_0$ .

## 2.2 Single Infection Between-Host Model

We consider the *simple infection (cross-immunity) model*. In this setting, hosts infected by a pathogen trait are fully protected against infection by any other trait, so that doubly infected host classes are absent.<sup>2</sup> A key modelling choice is that the virulence  $\alpha(p)$  and transmission rate  $\beta(p)$  corresponding to  $p$  are mechanistically related to the within-host equilibrium:

$$\alpha(p) = A(\hat{T}(p), \hat{T}^*(p), \hat{V}(p)), \quad \beta(p) = B(\hat{T}(p), \hat{T}^*(p), \hat{V}(p)),$$

where  $A, B \in \mathcal{C}^2(\mathbb{R}^3, \mathbb{R})$ .

**Definition 2.2** (Refined SI Model). For two traits  $p$  and  $q$ , we have the following ODE system:

$$\frac{dS}{dt} = b - \delta S - \beta(p)SI_p - \beta(q)SI_q, \quad (2.11)$$

$$\frac{dI_p}{dt} = -\delta I_p + \beta(p)SI_p - \alpha(p)I_p, \quad (2.12)$$

$$\frac{dI_q}{dt} = -\delta I_q + \beta(q)SI_q - \alpha(q)I_q, \quad (2.13)$$

where  $I_p$  and  $I_q$  are the numbers of the hosts infected with traits  $p$  and  $q$ .

*Remark.* Given the assumption of cross-immunity, we do not include dual-infection terms such as  $I_{pq}$ . The model also appeared in the introduction, with the difference that we have formulated it for two traits.

In the absence of trait  $q$ , i.e.,  $I_q = 0$ , the model is simplified to:

$$\frac{dS}{dt} = b - \delta S - \beta(p)SI_p, \quad (2.14)$$

$$\frac{dI_p}{dt} = \beta(p)SI_p - (\alpha(p) + \delta)I_p. \quad (2.15)$$

At the equilibrium,  $I_p = 0$  implies  $\frac{dI_p}{dt} = 0$ . By (2.14), we get the non-infectious steady state:

$$\hat{S}(p) = \frac{b}{\delta} =: S_0, \quad \hat{I}_p(p) = 0.$$

For the non-trivial case, by (2.15), we get

$$(\beta(p)S - (\alpha(p) + \delta))I_p = 0.$$

<sup>2</sup>This assumption is arguably inconsistent with allowing within-host selection. If a within-host mutant can take over, why can't an externally introduced competitor do so? Nevertheless, we keep it for the introduction of the superinfection model.

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For  $I_p > 0$ ,  $\beta(p)S - (\alpha(p) + \delta) = 0$  and

$$\hat{S}(p) = \frac{\alpha(p) + \delta}{\beta(p)}.$$

Plugging back to (2.14) gives

$$0 = b - \delta\hat{S}(p) - \beta(p)\hat{S}(p)I_p \implies \hat{I}_p(p) = \frac{b - \delta\hat{S}(p)}{\beta(p)\hat{S}(p)} = \frac{b}{\alpha(p) + \delta} - \frac{\delta}{\beta(p)}.$$

The biologically meaningful equilibrium requires  $\hat{S}(p) > 0$  and  $\hat{I}_p(p) > 0$ . Given that the transmission rate is positive,  $\hat{S}(p) > 0$ . For  $\hat{I}_p(p) > 0$ , we have

$$\frac{b}{\alpha(p) + \delta} - \frac{\delta}{\beta(p)} > 0 \iff \frac{b}{\delta} \cdot \frac{\beta(p)}{\alpha(p) + \delta} > 1.$$

**Definition 2.3** (Invasion Exponent). For mutant trait  $q$  in resident  $p$ , linearising (2.13) at  $I_q \approx 0$  gives

$$\frac{dI_q}{dt} \approx \beta(q)\hat{S}(p)I_q - (\alpha(q) + \delta)I_q,$$

we then define the *invasion exponent*:

$$s_p(q) := \beta(q)\hat{S}(p) - (\alpha(q) + \delta), \quad (2.16)$$

with

$$\hat{S}(p) = \frac{\alpha(p) + \delta}{\beta(p)}.$$

when

$$\mathcal{R}_0(p) := \frac{b}{\delta} \cdot \frac{\beta(p)}{\alpha(p) + \delta} > 1.$$

*Remark.* (i) The sign of  $s_p(q)$  determines whether  $q$  can invade:  $s_p(q) > 0$  implies growth from rarity, while  $s_p(q) < 0$  implies extinction.

(ii) To see the linearisation more religiously, write the two-trait single-infection model in vector form  $\dot{x} = F(x)$  with  $x = (S, I_p, I_q)^\top$ , and (2.13) becomes

$$F_3(S, I_p, I_q) = \beta(q)SI_q - (\alpha(q) + \delta)I_q.$$

Let  $\xi = (u, v, w)^\top := x - E_p$  be a small perturbation around the equilibrium  $E_p = (\hat{S}(p), \hat{I}_p(p), 0)$ , i.e.,

$$S = \hat{S}(p) + u, \quad I_p = \hat{I}_p(p) + v, \quad I_q = w.$$

The first-order Taylor expansion of  $F$  around  $E_p$  gives

$$\dot{\xi} = DF(E_p)\xi + \mathcal{O}(\|\xi\|^2) \approx DF(E_p)\xi.$$

Then the Jacobian in the  $I_q$ -direction satisfies

$$\left. \frac{\partial F_3}{\partial I_q} \right|_{E_p} = \beta(q)\hat{S}(p) - (\alpha(q) + \delta).$$

Therefore, we achieve the linearised equation.

The definition yields an optimisation principle: evolution tends to maximise  $\mathcal{R}_0(p)$ , or equivalently, to decrease the equilibrium  $\hat{S}(p)$ .

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## 2.3 Superinfection Model

The single-infection between-host model excludes the possibility of externally introduced strains replacing established strains within infected hosts. However, this conflicts with the requirement for within-host selection. Therefore, we decouple the within-host and between-host levels by allowing superinfection. Throughout, we assume:

- (i) Within-host dynamics reach equilibrium faster relative to between-host transmission.
- (ii) Mutations are rare on ecological timescales due to adaptive dynamics.
- (iii) Whenever a second trait is introduced into an already infected host, it either fails to take over or rapidly replaces the original trait, meaning that we assume an (almost) immediate trait replacement within the host.

**Definition 2.4** (Superinfection Function). Let  $p$  be the resident pathogen trait in a host, and  $q$  be the trait introduced into the same host through transmission. We define the *superinfection function*

$$\phi(p, q) := \psi \left( \hat{T}(p), \hat{T}(q) \right),$$

where  $\hat{T}(\cdot)$  is the equilibrium of uninfected target cells from the within-host model, and  $\psi: \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$  is a non-negative function.

The superinfection function describes the ability of trait  $q$  to take over a host infected by trait  $p$ . Biologically, if  $q$  is a worse (or equal) within-host competitor than  $p$ , i.e.,  $\hat{T}(p) < \hat{T}(q)$ , then  $q$  cannot establish in an already-infected host by  $p$ . We summarise this as the following restriction.

**Lemma 2.5.**  $\psi(x, y) = 0$  whenever  $y \geq x$ .

**Definition 2.6** (Superinfection Model). We largely retain the single-infection between-host model, with the key additional assumption being that infected hosts can be reinfected, after which one of the traits takes over. The system of *superinfection model* is given by:

$$\frac{dS}{dt} = b - \delta S - \beta(p)SI_p - \beta(q)SI_q, \quad (2.17)$$

$$\frac{dI_p}{dt} = -\delta I_p + \beta(p)SI_p - \alpha(p)I_p + \Phi(q, p)I_p I_q, \quad (2.18)$$

$$\frac{dI_q}{dt} = -\delta I_q + \beta(q)SI_q - \alpha(q)I_q + \Phi(p, q)I_p I_q, \quad (2.19)$$

where

$$\Phi(p, q) = \beta(q)\phi(p, q) - \beta(p)\phi(q, p).$$

*Remark.* The superinfection terms cancel each other out in  $\frac{d}{dt}(I_p + I_q)$ , thus superinfection merely redistributes infected hosts across different traits rather than directly inducing or eliminating infection.

Let trait  $p$  be resident and assume it persists at the endemic equilibrium  $E_p = (\hat{S}(p), \hat{I}_p, 0)$  in the absence of trait  $q$ . Linearising (2.19) around  $I_q \approx 0$  yields

$$\dot{I}_q \approx r_p(q) I_q,$$

where we have the invasion exponent:

$$\begin{aligned} r_p(q) &= \beta(q)\hat{S}(p) - (\alpha(q) + \delta) + \Phi(p, q)\hat{I}_p, \\ &= s_p(q) + \Phi(p, q)\hat{I}_p \\ &= s_p(q) + \hat{I}_p(\beta(q)\phi(p, q) - \beta(p)\phi(q, p)) \\ &= s_p(q) + \hat{I}_p \left( \beta(q)\psi \left( \hat{T}(p), \hat{T}(q) \right) - \beta(p)\psi \left( \hat{T}(q), \hat{T}(p) \right) \right), \end{aligned}$$

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where  $s_p(q)$  is exactly the invasion exponent we obtained in the single-infection model. For the equilibrium, we have

$$\hat{S}(p) = \frac{\alpha(p) + \delta}{\beta(p)}, \quad \hat{I}(p) = \frac{b}{\alpha(p) + \delta} - \frac{\delta}{\beta(p)} = \frac{b}{\beta(p)\hat{S}(p)} - \frac{\delta}{\beta(p)}. \quad (2.20)$$

The evolutionary consequences of superinfection critically depend on how  $\psi(x, y)$  approaches 0 as  $y \uparrow x$ . Depending on the behaviour, we can distinguish the following three cases (*Figure 1*).

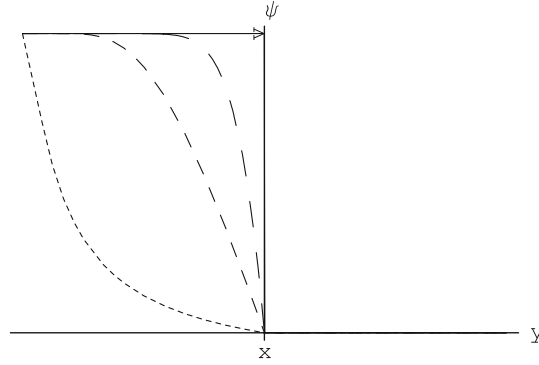


Figure 1: For  $y < x$ , the solid line represents the “jump” case, the two dashed lines in the middle represent the “mechanistic” case, and the bottom dashed line represents the “smooth” case.

### The “Jump” Case

We say  $\psi$  is the *jump* case if  $\psi$  has a jump discontinuity at the diagonal:

$$\lim_{y \uparrow x} \psi(x, y) = \psi_0(x) > 0, \quad \psi(x, x) = 0.$$

In this case, an arbitrarily small within-host advantage suffices to yield an  $\mathcal{O}(1)$  takeover probability, and hence within-host selection dominates the evolutionary outcome.

### The “Mechanistic” Case

We say  $\psi$  is the *mechanistic* case if  $\psi$  is continuous at  $y = x$  but has a one-sided slope discontinuity.

**Lemma 2.7.** *The one-sided limits exist:*

$$D(x) := \lim_{y \uparrow x} \frac{\psi(x, y)}{y - x} = -\lim_{y \downarrow x} \frac{\psi(y, x)}{y - x} \in (-\infty, 0).$$

So that locally

$$\psi(x, y) \approx D(x)(y - x), \quad y \uparrow x.$$

This case is the most pertinent to superinfection. Initially, the invading trait  $q$  is in small quantities and may go extinct due to stochasticity. We model it by a stochastic birth-death process. Assume the resident trait  $p$  is at its within-host equilibrium, so the invader trait  $q$  experiences a constant target-cell:

$$T = \hat{T}(p) = \frac{c}{k(\mathcal{B}_0(p) - 1)}. \quad (2.21)$$

The probability that a single free pathogen entering an uninfected target cell is

$$\frac{k\hat{T}(p)}{k\hat{T}(p) + c}.$$

\*\*\*\*\*

Otherwise, it is cleared with probability

$$\frac{c}{k\hat{T}(p) + c}.$$

Conditioned on successful entry, the infected cell's lifetime is exponentially distributed with rate  $\mu(q) + d$ , and it produces free pathogens according to a Poisson process with rate  $q$ . Integrating over all lifetimes yields an offspring distribution with probability generating function

$$G(z) = \frac{c}{k\hat{T}(p) + c} + \frac{k\hat{T}(p)}{k\hat{T}(p) + c} \cdot \frac{1}{1 + \mathcal{B}_0(q)(1 - z)}. \quad (2.22)$$

The extinction probability  $P_1(p, q)$  is the smallest solution of  $z = G(z)$ , combining with (2.21) and (2.22) gives

$$P_1(p, q) = \min \left\{ 1, \frac{c}{c + k\hat{T}(p)} + \frac{1}{\mathcal{B}_0(q)} \right\} = \min \left\{ 1, 1 - \frac{1}{\mathcal{B}_0(p)} + \frac{1}{\mathcal{B}_0(q)} \right\}.$$

Hence, the survival (establishment) probability of a single introduced particle is

$$\phi_1(p, q) = \max \left\{ 0, \frac{1}{\mathcal{B}_0(p)} - \frac{1}{\mathcal{B}_0(q)} \right\}.$$

If the transmission dose is  $n$  with independent initial lineages, the probability that at least one lineage survives is

$$\begin{aligned} \phi_n(p, q) &= \begin{cases} 1 - P_1^n(p, q), & \mathcal{B}_0(p) < \mathcal{B}_0(q) \\ 0, & \text{otherwise} \end{cases} \\ &= \begin{cases} 1 - \left( 1 - \frac{k\hat{T}(p)}{c + k\hat{T}(p)} + \frac{k\hat{T}(q)}{c + k\hat{T}(q)} \right)^n, & \hat{T}(q) < \hat{T}(p) \\ 0, & \text{otherwise} \end{cases} \\ &\sim \begin{cases} -n \frac{kc}{(c + kp)^2} (q - p), & q \uparrow p \\ 0, & q \downarrow p \end{cases}. \end{aligned}$$

Therefore, the resulting takeover probability  $\phi(p, q)$  is continuous but not differentiable at  $q = p$ , which yields the mechanistic case.

### The ‘Smooth’ Case

We say  $\psi$  is the *smooth* case if it is differentiable at  $y = x$ . By *Lemma 2.5*, differentiability forces the derivative at the diagonal to vanish:

$$\left. \frac{\partial \psi}{\partial y} \right|_{(x, x)} = 0,$$

and therefore  $\psi(x, y) = o(|y - x|)$  as  $y \uparrow x$ . In this case, superinfection exhibits no first-order effect on the selection gradient, and the optimal strategy aligns with that we observed in the single-infection model.

## 3 Main Results

In this section, we analyse how superinfection alters singular strategies and how the loss of one-dimensional optimisation principles leads to evolutionarily stable coexistence.

### 3.1 Virulence

Within one host, evolution minimises the equilibrium level  $\hat{T}(p)$ . We denote by  $p^*$  a (locally) continuously stable strategy (CSS), i.e., a local attractor under the adaptive dynamics. In the single-infection model, selection at the host population level operates through the susceptible population. The evolutionary dynamics tend to minimise  $\hat{S}(p)$ , and we denote the resulting CSS as  $p^\bullet$  characterised by

$$\left. \frac{\partial s_{p^\bullet}(q)}{\partial q} \right|_{q=p^\bullet} = 0, \quad \left. \frac{\partial^2 s_{p^\bullet}(q)}{\partial q^2} \right|_{q=p^\bullet} < 0.$$

When superinfection is allowed, and the superinfection function is mechanistic, the invasion exponent becomes

$$r_p(q) = \beta(q)\hat{S}(p) - (\alpha(q) + \delta) + \Phi(p, q)\hat{I}_p.$$

And the selection gradient for monomorphic evolution is given by

$$\begin{aligned} \left. \frac{\partial r_p(q)}{\partial q} \right|_{q=p} &= \left. \frac{\partial s_p(q)}{\partial q} \right|_{q=p} + \beta(p)D(\hat{T}(p))\hat{I}(p) \left. \frac{d\hat{T}}{dq} \right|_{q=p} \\ &= -\beta(p) \left[ \left. \frac{d\hat{S}}{dq} \right|_{q=p} - D(\hat{T}(p))\hat{I}(p) \left. \frac{d\hat{T}}{dq} \right|_{q=p} \right], \end{aligned}$$

where  $D(\cdot) \in (-\infty, 0)$ . Since  $\hat{I}(p) > 0$  at endemic equilibrium, the term  $-D(\hat{T}(p))\hat{I}(p)$  is strictly positive, so it can only vanish in regions where  $\hat{S}'(p)$  and  $\hat{T}'(p)$  have opposite signs. A singular strategy  $p^{\bullet*}$  for the monomorphic superinfection dynamics is given by

$$\left. \frac{\partial r_p(q)}{\partial q} \right|_{q=p^{\bullet*}} = 0,$$

and it is *convergence stable* if the adaptive dynamics flows towards  $p^{\bullet*}$ .

Assume that  $p^*$  and  $p^\bullet$  are (locally) unique and, without loss of generality,  $p^\bullet < p^*$ . Then, by the uniqueness of the minimisers, we have

$$\begin{cases} \hat{S}'(p) < 0, & p < p^\bullet \\ \hat{S}'(p) > 0, & p > p^\bullet \end{cases}, \quad \begin{cases} \hat{T}'(p) < 0, & p < p^* \\ \hat{T}'(p) > 0, & p > p^* \end{cases}.$$

In particular, for all  $p \in (p^\bullet, p^*)$  we have  $\hat{S}'(p) > 0$  but  $\hat{T}'(p) < 0$ . Therefore, when the within-host CSS  $p^*$  and the between-host CSS  $p^\bullet$  are unique, the convergence stable strategy  $p^{\bullet*}$  lies in between (Figure 2).

When  $\hat{S}'(p) = 0$ ,

$$\text{sign} \left. \frac{\partial r_p(q)}{\partial q} \right|_{q=p} = -\text{sign} \left. \frac{d\hat{T}}{dq} \right|_{q=p}.$$

Whenever  $\hat{T}'(p^\bullet) < 0$ , mutants with slightly larger  $q > p^\bullet$  can invade, and the trait is driven away from  $p^\bullet$  to  $p^{\bullet*}$ , in the direction of decreasing  $\hat{T}$ .

Finally, virulence is mechanistically linked to within-host equilibrium via

$$\alpha(p) = A(\hat{T}(p), \hat{T}^*(p), \hat{V}(p)).$$

For biologically reasonable choices of  $A$ , moving in the direction of decreasing  $\hat{T}$  leads to an increase in  $\alpha$ . Therefore, superinfection can increase the evolutionarily selected level of virulence.

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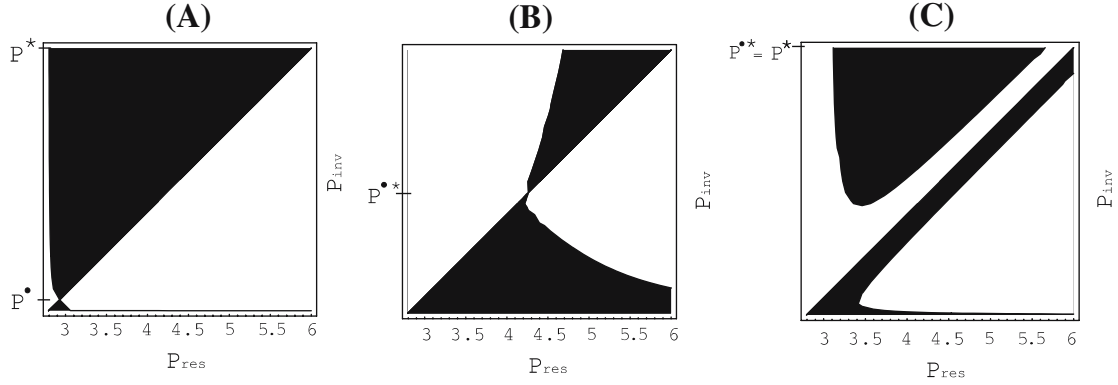


Figure 2: Pairwise invasibility plots where (A) “smooth” case; (B) “mechanistic” case; (C) “jump” case.  $p^* = p_{\max}$ .  $p_{\text{res}}$  denotes the white region, and  $p_{\text{inv}}$  denotes the black region.

### 3.2 Evolutionarily Stable Coexistence

For single-infection models, the pathogen resides in a one-dimensional environment, and thus its adaptive dynamics typically follow optimisation principles. Permitting superinfection disrupts this structure: the environment is no longer one-dimensional, and we should not expect a single optimal quantity [2].

Consequently, pairwise invasibility plots typically lose their partial symmetry (*Figure 2, (B)* “mechanistic” case and *(C)* “jump” case lack skew symmetry).<sup>3</sup> This implies the existence of trait pairs  $(p, q)$  for which

$$r_p(q) > 0, \quad r_q(p) > 0,$$

which is called *mutual invasibility*, yielding *protected dimorphism* on the epidemiological time scale (*Figure 3*).

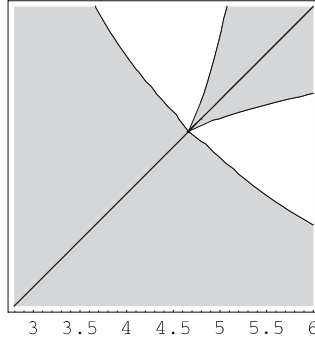


Figure 3: White areas correspond to the regions in  $(p, q)$  space in which  $p$  and  $q$  are mutually invadable. Grey regions are the areas where  $r_p(q)$  and  $r_q(p)$  have opposite signs.

Assume traits  $p_1$  and  $p_2$  are both resident, and consider a mutant  $q$  introduced at low frequency. We have the invasion exponent

$$r_{p_1, p_2}(q) = \beta(q)\hat{S}(p_1, p_2) - (\alpha(q) + \delta) + \Phi(p_1, q)\hat{I}_{p_1}(p_1, p_2) + \Phi(p_2, q)\hat{I}_{p_2}(p_1, p_2), \quad (3.1)$$

where  $\hat{S}(p_1, p_2)$  is the susceptible equilibrium under the two-trait resident state, and  $\hat{I}_{p_i}(p_1, p_2)$  are the corresponding equilibrium prevalences. Then we rewrite (2.17)-(2.19) as:

$$\frac{dS}{dt} = b - \delta S - \beta_1 S I_1 - \beta_2 S I_2, \quad (3.2)$$

<sup>3</sup>Skew symmetry means invariance under reflection across the main diagonal, together with an interchange of black and white.

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$$\frac{dI_1}{dt} = \beta_1 S I_1 - (\alpha_1 + \delta) I_1 + \Phi(p_2, p_1) I_1 I_2 = I_1 (\beta_1 S - (\alpha_1 + \delta) + \Phi_{21} I_2), \quad (3.3)$$

$$\frac{dI_2}{dt} = \beta_2 S I_2 - (\alpha_2 + \delta) I_2 + \Phi(p_1, p_2) I_1 I_2 = I_2 (\beta_2 S - (\alpha_2 + \delta) + \Phi_{12} I_1). \quad (3.4)$$

Let  $\dot{I}_1 = \dot{I}_2 = 0$ , we have

$$0 = \beta_1 \hat{S} - (\alpha_1 + \delta) + \Phi_{21} \hat{I}_2, \quad (3.5)$$

$$0 = \beta_2 \hat{S} - (\alpha_2 + \delta) + \Phi_{12} \hat{I}_1. \quad (3.6)$$

Observe that

$$\Phi(q, p) = \beta(p) \phi(q, p) - \beta(q) \phi(p, q) = -\Phi(p, q),$$

which implies  $\Phi_{21} = -\Phi_{12}$ . Then by (3.5)-(3.6) and (2.20), we compute

$$\begin{aligned} \hat{I}_1 &= \frac{\alpha_2 + \delta - \beta_2 \hat{S}}{\Phi} = \frac{\beta_2 \hat{S}(p_2) - \beta_2 \hat{S}}{\Phi} = \frac{\beta_2}{\Phi} (\hat{S}(p_2) - \hat{S}), \\ \hat{I}_2 &= \frac{\beta_1 \hat{S} - (\alpha_1 + \delta)}{\Phi} = \frac{\beta_1 \hat{S} - \beta_1 \hat{S}(p_1)}{\Phi} = \frac{\beta_1}{\Phi} (\hat{S} - \hat{S}(p_1)), \end{aligned}$$

which gives (3.8) and (3.9). By (3.2),

$$\begin{aligned} b &= \hat{S} (\delta + \beta_1 \hat{I}_1 + \beta_2 \hat{I}_2) \\ &= \hat{S} \left( \delta + \frac{\beta_1 \beta_2}{\Phi} (\hat{S}(p_2) - \hat{S}) + \frac{\beta_1 \beta_2}{\Phi} (\hat{S} - \hat{S}(p_1)) \right) \\ &= \hat{S} \left( \delta + \frac{\beta_1 \beta_2}{\Phi} (\hat{S}(p_2) - \hat{S}(p_1)) \right). \end{aligned}$$

Then

$$\hat{S} = \frac{b}{\delta + \frac{\beta_1 \beta_2}{\Phi} (\hat{S}(p_2) - \hat{S}(p_1))} = \frac{b \Phi}{\delta \Phi + \beta_1 \beta_2 (\hat{S}(p_2) - \hat{S}(p_1))},$$

which gives (3.7). Therefore, we have computed that:

$$\hat{S}(p_1, p_2) = \frac{b \Phi(p_1, p_2)}{\delta \Phi(p_1, p_2) + \beta(p_1) \beta(p_2) (\hat{S}(p_2) - \hat{S}(p_1))}, \quad (3.7)$$

$$\hat{I}_{p_1}(p_1, p_2) = \frac{\beta(p_2)}{\Phi(p_1, p_2)} (\hat{S}(p_2) - \hat{S}(p_1, p_2)), \quad (3.8)$$

$$\hat{I}_{p_2}(p_1, p_2) = \frac{\beta(p_1)}{\Phi(p_1, p_2)} (\hat{S}(p_1, p_2) - \hat{S}(p_1)), \quad (3.9)$$

where  $\hat{S}(p) = \frac{\alpha(p) + \delta}{\beta(p)}$ . Without loss of generality, we assume  $\hat{T}(p_2) < \hat{T}(p_1)$  so that  $p_2$  wins within-host competition whenever superinfection succeeds. By *Lemma 2.5*,

$$\Phi(p_1, p_2) = \beta(p_2) \phi(p_1, p_2), \quad \phi(p_2, p_1) = 0.$$

By (3.8) and (3.9), feasibility of coexistence is equivalent to

$$\hat{S}(p_1) < \hat{S}(p_1, p_2) < \hat{S}(p_2), \quad (3.10)$$

where  $p_1$  can invade the  $p_2$ -only equilibrium when  $\hat{S}(p_1, p_2) < \hat{S}(p_2)$ , while  $p_2$  can invade the  $p_1$ -only equilibrium when  $\hat{S}(p_1) < \hat{S}(p_1, p_2)$ . Hence (3.10) is exactly the *mutual invasibility* condition.

Protected dimorphisms can appear as transients (*converging dimorphism*) near a CSS that is uninvable. Once dimorphism is established, the monomorphic selection gradient  $\left. \frac{\partial r_p(q)}{\partial q} \right|_{q=p}$  is no longer sufficient. Instead, we can define the *internal singular coalition*  $(p_1, p_2)$  such that

$$\left. \frac{\partial}{\partial q} r_{p_1, p_2}(q) \right|_{q=p_1} = 0, \quad \left. \frac{\partial}{\partial q} r_{p_1, p_2}(q) \right|_{q=p_2} = 0. \quad (3.11)$$

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Geometrically, each equation in (3.11) represents an *isocline* in the  $(p_1, p_2)$ -plane. The intersection in the coexistence region corresponds to a candidate *singular coalition*, as demonstrated by a numerical example. A singular coalition is *evolutionarily stable* (uninvadable) if

$$r_{p_1, p_2}(q) < 0, \quad \forall q \notin \{p_1, p_2\}.$$

In the mechanistic case, the monomorphic attractor  $p^{\bullet*}$  can become a *branching point*, which means convergence stable but invadable. It is because the superinfection function is continuous but not differentiable on the diagonal, so the local curvature of  $q \mapsto r_p(q)$  can be one-sided and even asymmetric.

**Example 3.1** (Numerical Procedure). We simulate evolution using a trait-substitution sequence (TSS) with parameters:

- Within-Host:  $\lambda = 1$ ,  $k = 10$ ,  $c = 0.1$ ,  $d = 2$ .
- Between-Host:  $b = 1$ ,  $\delta = 0.1$ .
- Further:

$$\mu(p) = \frac{p}{p+1}, \quad \alpha(p) = \frac{k}{\lambda} \hat{V}(p), \quad \beta(p) = \hat{V}(p) + \hat{T}^*(p),$$

where  $(\hat{T}(p), \hat{T}^*(p), \hat{V}(p))$  is the within-host equilibrium.

For the superinfection function  $\psi_n$  we set  $n = 200$ , thus

$$\phi(p, q) = \psi_{200}(\hat{T}(p), \hat{T}(q)).$$

We repeatedly propose a small mutant  $q = p^{(m)} + \Delta$  and compute its invasion exponent. If  $r_{p^{(m)}}(q) > 0$ , we accept substitution and set  $p^{(m+1)} = q$ ; otherwise we keep  $p^{(m+1)} = p^{(m)}$ . When the reinfection dose parameter is large, the monomorphic attractor  $p^{\bullet*} \approx 4.69345$  becomes an (asymmetric) branching point, which is invadable from above but not from below. After branching, we compute the ecological coexistence equilibrium  $(\hat{S}(p_1, p_2), \hat{I}_{p_1}(p_1, p_2), \hat{I}_{p_2}(p_1, p_2))$  by (3.7)-(3.9). A mutant  $q = p_i + \Delta$  replaces its parent branch if  $r_{p_1, p_2}(q) > 0$ , recall (3.1).

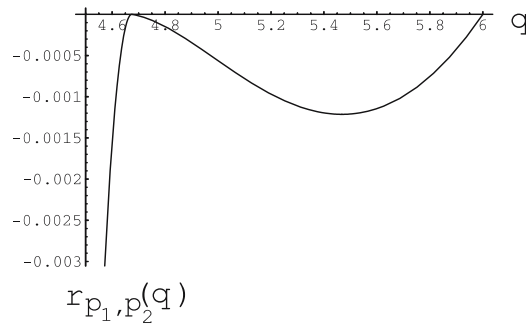


Figure 4: The graph of  $r_{4.675, 6}(q)$ .

Simulations suggest that the dimorphic TSS converges to *evolutionarily stable coalition*  $(p_1, p_2) \approx (4.675, 6)$ , where the upper branch hits the bound  $p_{\max} = 6$ . The negativity for all  $q \notin \{4.675, 6\}$  indicates the uninvadability of the coalition.

In conclusion, superinfection can turn  $p^{\bullet*}$  into a branching point, leading to evolutionarily stable coexistence of different strains.

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## 4 Conclusion

This essay investigates how superinfection changes pathogen evolution. We have mainly arrived at the following two results:

- Superinfection can promote virulence.
- Superinfection can generate coexistence not only on epidemiological time scales (protected dimorphism), but also evolutionarily stable coexistence of different strains.

Such results are suitable for chronic infections that require reinfection, such as HIV and Hepatitis B/C. Although our assumptions constrain the scope of the findings, we may still employ the evolutionary dynamics framework to investigate SEI and SIR models with different incidence rates by incorporating recovery and immunity.

Notice that several simplifying assumptions are made in the models. Rare mutations and the between-host model are SI with lifelong infection and no immune clearance. Within-host competition is target-cell limited and excludes long-term coinfection. Moreover, superinfection is idealised as (almost) instantaneous replacement. These constitute the principal limitations of this study.

As an ending remark, we primarily referred to Barbara Boldin and Odo Diekmann's article [3], with all figures and key content originating from this source. We also consulted our textbook [4] as some essential background knowledge.

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